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Supervisor: María Camila Atehortúa Cardona

Trade Salience and United Nations General Assembly (UNGA) Voting Similarity: Evidence from the Colombia–United States Dyad

Santiago Calderón Angarita

Matrikelnummer: 12515310

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1 Introduction

A longstanding question in international relations concerns whether economic ties between states shape their political behavior in multilateral forums (Barbieri 1996). If trade creates interdependence, and if that interdependence generates incentives for political cooperation (Keohane 1984), then states with stronger commercial relationships might be expected to align more closely on questions of international politics. This expectation is particularly plausible in asymmetric dyads, where the weaker partner depends heavily on the stronger one and may face implicit or explicit pressure to conform to its preferences.

This paper examines the relationship between trade salience and political alignment in the context of the Colombia–United States dyad. The research question is: how does the salience of bilateral trade relate to voting similarity in the United Nations General Assembly (UNGA)? Colombia represents an analytically interesting case. It is one of the United States' most significant partners in South America and has historically maintained an asymmetric relationship with the US across economic, political, and security dimensions. A country in this structural position might be expected to align its multilateral behavior with that of the hegemon, particularly when trade dependence is high.

Drawing on the argument that trade dependence can function as a source of political leverage (Hirschman 1945), and on the liberal proposition that state preferences shaped by economic interests translate into foreign policy behavior (Moravcsik 1997), the expectation motivating this analysis is that higher trade salience is positively associated with greater voting similarity at the UNGA. This expectation is further supported by the observation that asymmetric trade relationships create structural conditions for coercion that symmetric ones do not (Barbieri 1996). The dependent variable (Y) is dyad–year voting similarity, operationalized as the share of roll-call votes in which Colombia and the United States cast identical votes. The independent variable (X) is trade salience, measured as the share of Colombia's total trade conducted with the United States. The analysis is conducted at the dyadic level of analysis, as the focus is on patterns of political similarity within a bilateral relationship rather than the behavior of individual states in isolation.

The remainder of this paper is structured as follows. Section 2 presents the research design, including sample construction, operationalization of both variables, and descriptive statistics. Section 3 presents the bivariate analysis. Section 4 discusses the findings and their limitations. Section 5 concludes.

2 Research Design

This section describes the sample construction and data selection, followed by the operationalization and descriptive analysis of the dependent and independent variables.

2.1 Sample Construction and Data Selection

The unit of analysis is the dyad–year, defined as the Colombia–United States bilateral relationship observed annually. The sample is longitudinal, covering the period 1991–2019. The lower bound is determined by data availability: annual bilateral trade data from the World Bank Integrated Trade Solution (WITS) portal is available for Colombia–US trade from 1991 onward. The upper bound is determined by the voting similarity data, which extends to 2019 in the version of the `unvotes` R package used here.

Two data sources are merged by year to construct the final dataset. UNGA voting similarity is derived from the `unvotes` package (Robinson and Gohel 2021), which records roll-call votes for all UN member states. Trade salience is constructed from two sources: annual Colombia–US bilateral trade figures (exports and imports) from WITS, and Colombia's total trade from the World Bank API, accessed via the `wbstats` R package. Years are retained only when both non-missing voting similarity and non-missing trade data are available. The final sample consists of $n = 29$ dyad–years, covering every year from 1991 to 2019 without gaps.

2.2 Operationalization of Y: UNGA Voting Similarity

The dependent variable is UNGA voting similarity between Colombia and the United States. It is operationalized as the share of roll-call votes in a given year in which both countries cast identical votes, conditional on both having a recorded vote (yes, no, or abstain). Similarity is calculated based on identical voting positions across all recorded votes, including instances where both states chose to abstain. When both countries abstain on the same resolution, this is treated as a shared diplomatic preference for neutrality and counted as agreement. A vote is only excluded when one country has a recorded position and the other does not.

Voting similarity at the UNGA is an established measure of political alignment in the quantitative international relations literature. When two states consistently vote in the same direction across a wide range of issues, this signals ideological proximity, shared interests, or political coordination. The measure is continuous, bounded between 0 and 1, and constitutes a ratio-level variable. Since both bounds are meaningful (0 indicates no identical votes; 1 indicates perfect agreement), measures of central tendency and dispersion are appropriate for its description.

Over the 1991–2019 period, Colombia–US voting similarity exhibits a mean of 0.217 and a median of 0.221, indicating a low and relatively symmetric distribution. The standard deviation is 0.092, the minimum is 0.053, and the maximum is 0.450.

To provide historical context for these values, Figure 1 shows the full trend from 1946 to 2019. The rationale for including data prior to the study period is discussed in Section 4.

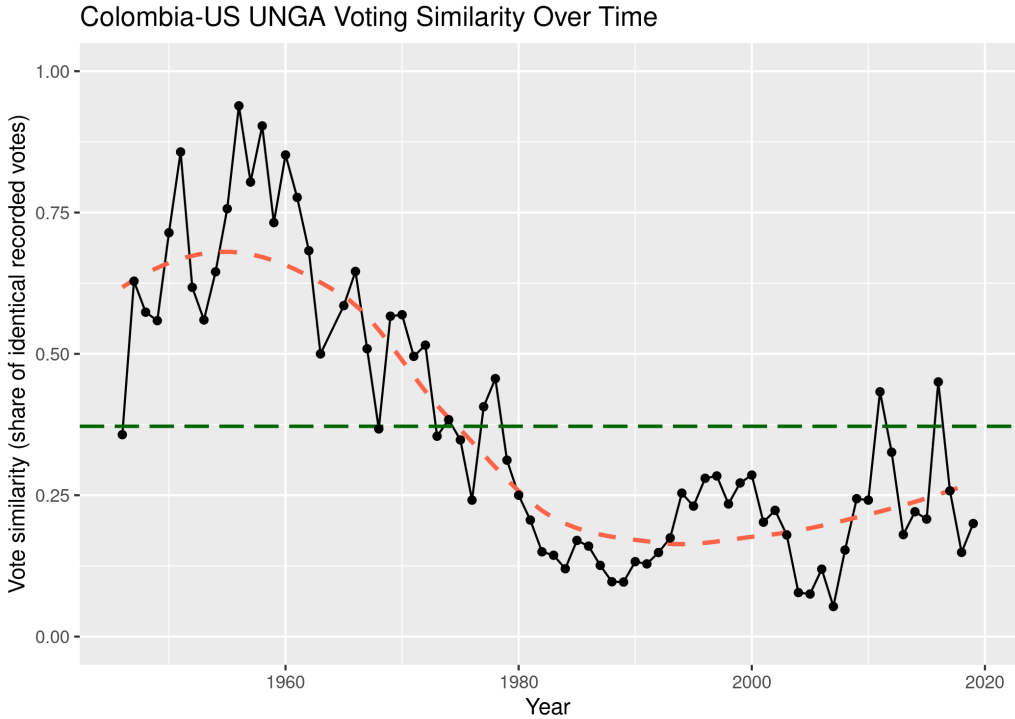

Figure 1: Colombia–US UNGA Voting Similarity, 1946–2019.
Note: Each point represents one dyad-year. The dashed red curve shows the general trend in the data. The dashed green horizontal line indicates the period mean (0.37).

The series makes clear that the low level of alignment observed during the study period is not a recent anomaly but the result of a long historical decline. Figure 2 zooms in on the study period 1991–2019. No strong upward or downward trend is visible, though notable spikes occur in 2011 and 2016, reaching similarity scores of approximately 0.43 and 0.45 respectively. These upward deviations likely reflect transient political developments in the bilateral relationship rather than structural changes in alignment; this point is taken up further in Section 4.

Before interpreting these scores, it is worth considering their reliability. A similarity score is a sample mean computed over the roll-call votes in a given year, and like any mean, its precision increases with the number of observations on which it is based. In years with few paired votes, the score has higher variance and is more sensitive to individual observations — a single unusual vote can shift the annual figure substantially. Figure 3 addresses this by encoding the number of jointly-recorded votes as the color of each point — darker points mean more votes, and therefore more reliable yearly estimates.

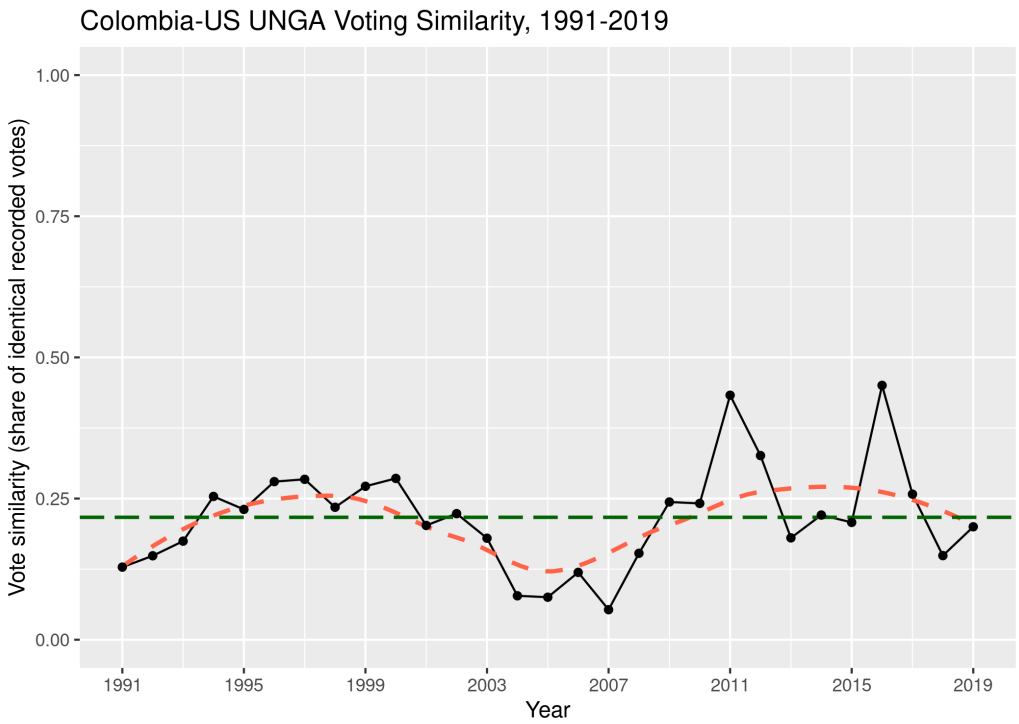

Figure 2: Colombia–US UNGA Voting Similarity, 1991–2019.

Note: Each point represents one dyad-year. The dashed red curve shows the general trend in the data. The dashed green horizontal line indicates the period mean (0.22).

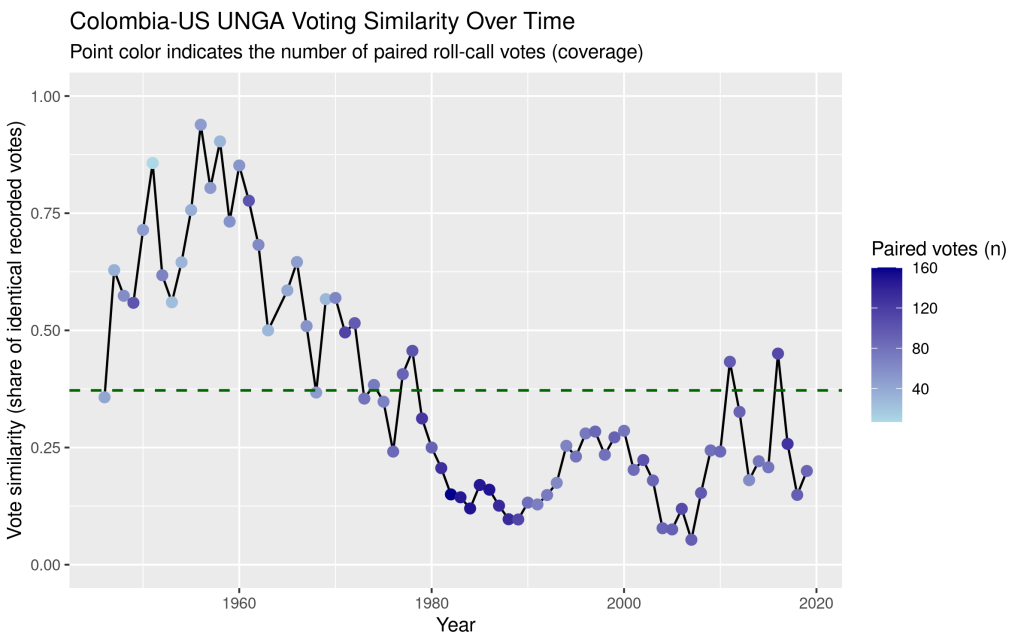

Figure 3: Colombia–US UNGA Voting Similarity and Vote Coverage, 1946–2019.

Note: Each point represents one dyad-year. The color indicates how many votes that year's similarity score is based on — darker blue means more votes, lighter blue means fewer. Years with more votes produce more reliable similarity scores.

The study period consistently shows high coverage (80–130 paired votes per year), meaning the yearly similarity scores are based on enough votes to be trustworthy. Finally, Figure 4 presents the distribution of voting similarity over the study period as a box plot, summarizing the central tendency and spread of Y in a single view.

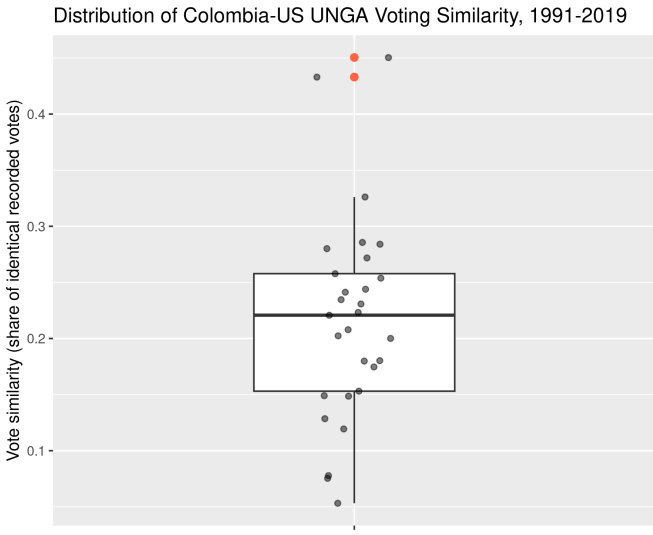

Figure 4: Distribution of Colombia–US UNGA Voting Similarity, 1991–2019.
Note: The box displays the interquartile range; the horizontal line indicates the median. Grey points show individual yearly observations. Red points are statistical outliers (2011 and 2016), which correspond to the spikes visible in Figure 2 and are discussed in Section 4.

The interquartile range is narrow (approximately 0.153 to 0.258), with two upward outliers corresponding to the spikes visible in Figure 2. A summary of the descriptive statistics is provided in Table 1 at the end of this section.

2.3 Operationalization of X: Trade Salience

The independent variable is trade salience, defined as the share of Colombia's total trade conducted with the United States in a given year. It is constructed as:

$$\text{Trade salience}_t = \frac{\text{Colombia–US bilateral trade}_t}{\text{Colombia total trade}_t}$$

where bilateral trade is the sum of Colombian exports to and imports from the United States in year t, measured in nominal US dollars at the prices of that year. Total trade is the sum of Colombia's total merchandise exports and imports in the same year, expressed in the same units.

This operationalization is preferred over the raw bilateral trade value in US dollars because it normalizes the bilateral relationship to Colombia's overall trade exposure. A large bilateral trade volume means little if Colombia trades extensively with many other

partners; what matters for the argument about dependence and leverage is the share of Colombia's trade that is concentrated in the US relationship. The use of both exports and imports, rather than one direction alone, captures the full bilateral commercial relationship.

Two data sources are required because no single source provides both measures in a consistent annual format for this dyad and period. Bilateral trade data are drawn from the World Bank WITS portal, which records annual commodity trade flows between country pairs. Colombia's total trade is obtained from the World Bank Development Indicators via the `wbstats` R package, using the standard exports and imports indicators.

Trade salience is a continuous ratio-level variable bounded between 0 and 1. Over the study period, it exhibits a mean of 0.341 and a median of 0.345, indicating that the US accounted for approximately one third of Colombia's total trade on average. The standard deviation is 0.045, the minimum is 0.263, and the maximum is 0.441.

Figure 5 displays the evolution of trade salience over the study period. The dashed line indicates the period mean (0.341) and the shaded band spans one standard deviation above and below it (0.296 to 0.386). Years falling outside the band deviate more than one standard deviation from the mean, signaling observations that are unusually high or low relative to the typical level of trade salience in the period.

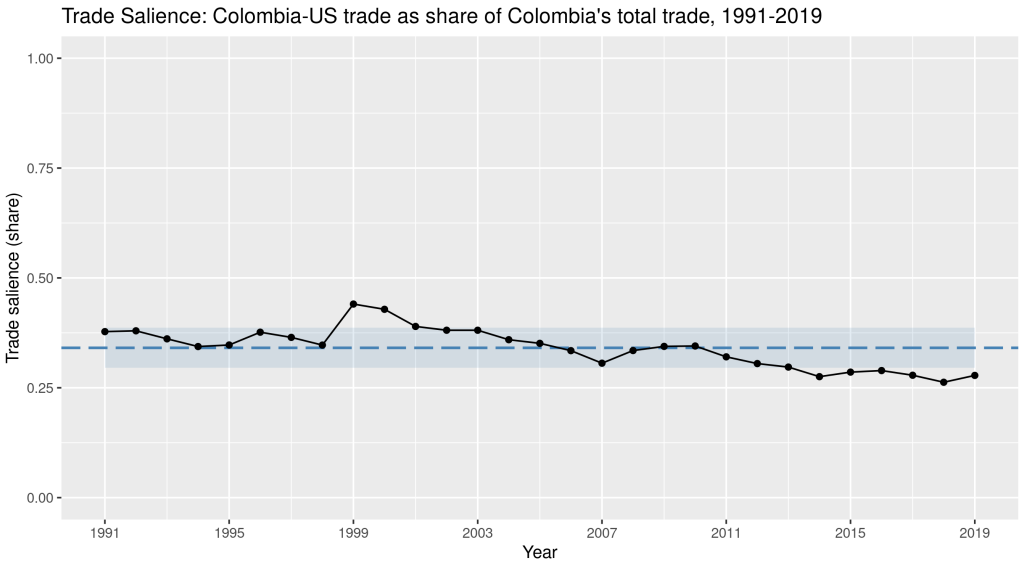

Figure 5: Colombia–US Trade Salience, 1991–2019.

Note: Trade salience is measured as Colombia–US bilateral trade (exports plus imports) as a share of Colombia's total merchandise trade. The dashed blue line indicates the period mean; the shaded band covers one standard deviation above and below the mean.

A gradual downward trend is visible: trade salience was highest in the late 1990s (peaking around 0.44 in 1999–2000) and declined steadily to approximately 0.26 by 2019, reflecting Colombia's trade diversification over the period. Figure 6 presents the same data as a box plot to show the overall distribution more clearly.

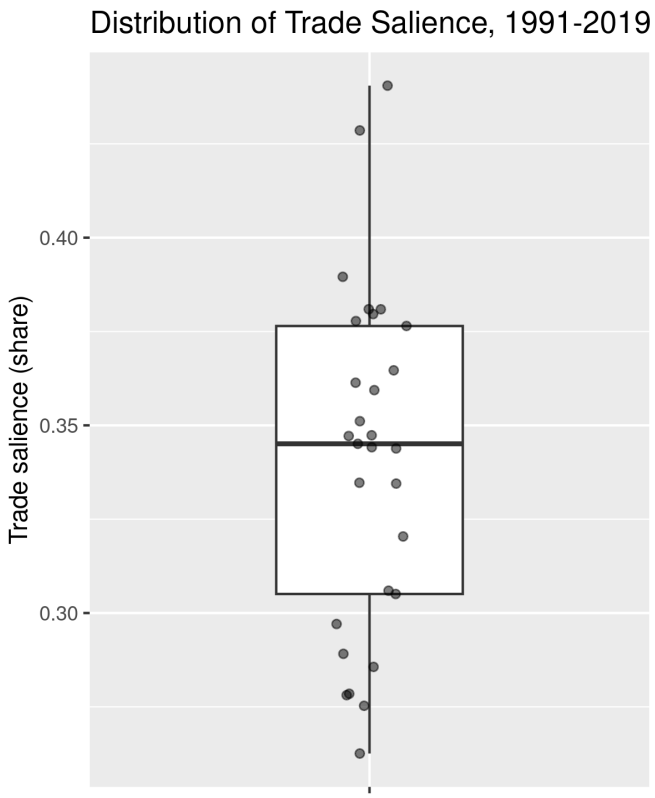

Figure 6: Distribution of Colombia–US Trade Salience, 1991–2019.

Note: The box displays the interquartile range; the horizontal line indicates the median. Grey points show individual yearly observations.

The distribution is compact and approximately symmetric with no extreme outliers, confirming that trade salience varied within a relatively narrow range despite the downward trend. Table 1 summarizes the descriptive statistics for both variables.

Table 1: Descriptive Statistics, 1991–2019 ($n = 29$)

Variable	Mean	Median	SD	Min	Max
UNGA voting similarity	0.217	0.221	0.092	0.053	0.450
Trade salience	0.341	0.345	0.045	0.263	0.441

Note: Both variables are measured as shares on a 0–1 scale. Voting similarity is the share of identical recorded roll-call votes in a given year. Trade salience is Colombia–US bilateral trade as a share of Colombia's total trade.

3 Bivariate Analysis

Both trade salience and voting similarity are continuous ratio-level variables. The appropriate method to assess their bivariate relationship is the Pearson product-moment correlation coefficient, which measures the direction and linear strength of association between two interval or ratio variables (Halperin and Heath 2020). A scatter plot is used to visualize the joint distribution.

Figure 7 presents the scatter plot of trade salience (X-axis) and voting similarity (Y-axis) for the 29 dyad–years in the sample. Each point represents one year.

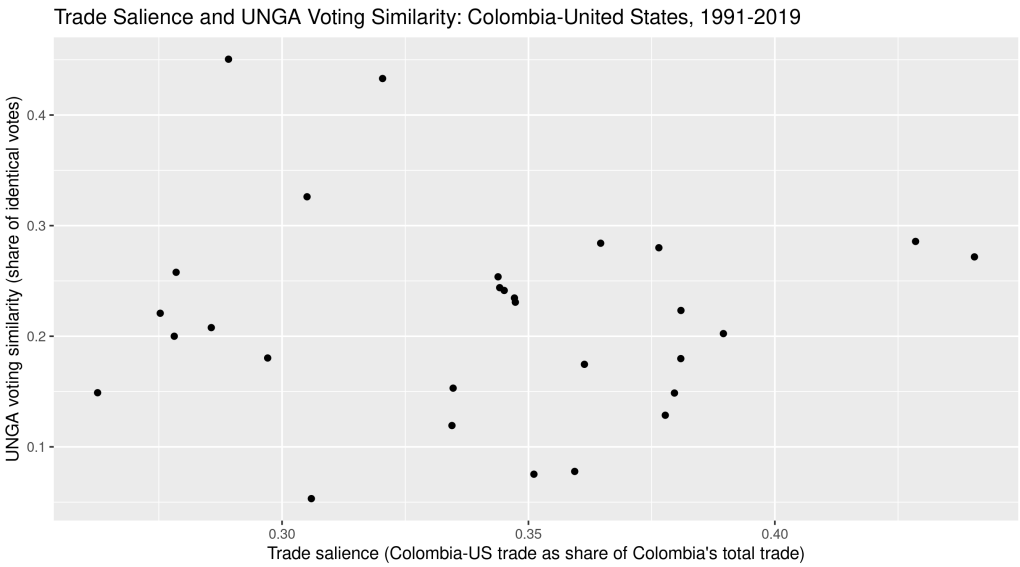

Figure 7: Trade Salience and UNGA Voting Similarity: Colombia–United States, 1991–2019.
Note: Each point represents one dyad-year. The Pearson correlation between trade salience and voting similarity is $r = -0.042$.

No discernible pattern is visible: the points are scattered without any systematic upward or downward tendency. Years with high trade salience are not consistently associated with higher or lower voting similarity, and the distribution of similarity scores appears roughly uniform across the range of observed salience values.

The Pearson correlation between trade salience and voting similarity is $r = -0.042$. This value is substantively negligible and in the opposite direction to the hypothesis. It indicates that, in this sample, there is essentially no linear association between the share of Colombia's trade conducted with the United States and the degree to which Colombia voted in alignment with the United States at the UNGA.

Figure 8 places both variables side by side as box plots, making it easier to compare their distributions directly.

Trade salience is centered higher and more compactly distributed than voting similar-

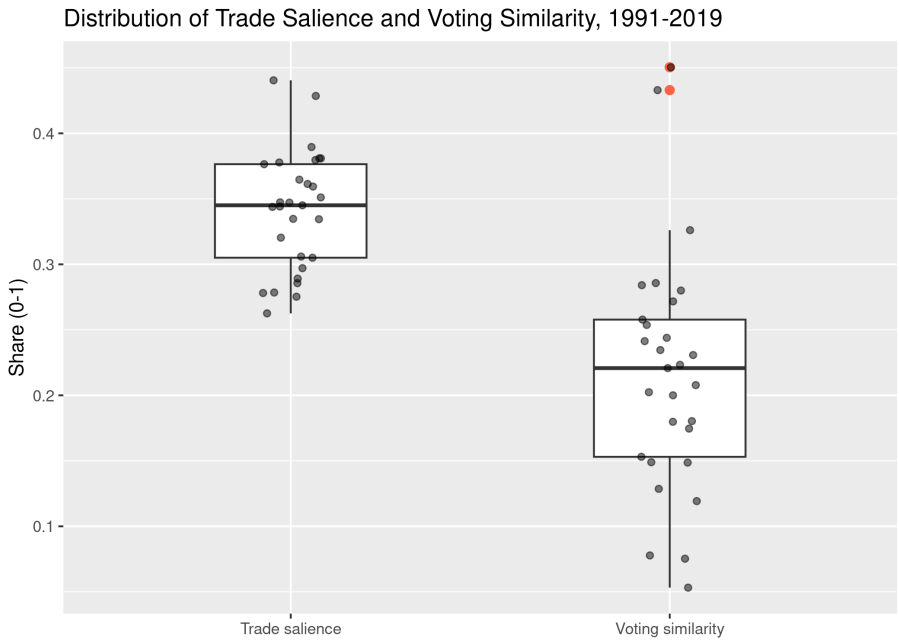

Figure 8: Distribution of Trade Salience and Voting Similarity, Colombia–United States, 1991–2019.

Note: The box displays the interquartile range; the horizontal line indicates the median. Red points are statistical outliers. Grey points show individual yearly observations. Both variables are measured as shares on a 0–1 scale.

ity. The two variables have different central tendencies and spreads, with no evident correspondence between their yearly values.

4 Discussion

The hypothesis stated that higher trade salience would be positively associated with greater Colombia–US voting similarity at the UNGA. The empirical analysis does not corroborate this expectation. The Pearson correlation of $r = -0.042$ is negligible in magnitude and marginally negative in direction, indicating that the data provide no evidence of a positive linear relationship between the two variables over the 1991–2019 period. This constitutes a null finding, which is a substantively informative result in itself (Halperin and Heath 2020).

Several explanations for this null result are worth considering. First, the measure of voting similarity used here does not distinguish between issue areas. UNGA votes cover a wide range of topics — human rights, disarmament, decolonization, development, and procedural matters — and it is possible that Colombia–US alignment varies substantially across these categories in ways that aggregate similarity scores obscure. A more refined analysis that weights or filters votes by topic could reveal patterns invisible at the aggregate level.

Second, the study period coincides with a series of political and security developments in the Colombia–US relationship that operated independently of trade. Colombia was among the largest recipients of US foreign assistance during this period, with security and counternarcotics aid under Plan Colombia from 2000 onward dwarfing the commercial relationship as the primary channel of bilateral engagement. The period also includes the Uribe presidency (2002–2010) and its particularly close alignment with US security priorities, the subsequent Santos administration (2010–2018), which pursued a more independent foreign policy, most notably through the FARC peace negotiations conducted with international guarantors outside US influence, and the conclusion of the peace process with the FARC in 2016. These shifts in the bilateral political and security relationship likely produced variation in voting similarity that is unrelated to trade dependence, and represent confounding factors that a bivariate analysis of trade and voting cannot account for.

Third, the time series of trade salience shows a steady structural decline over the period — from approximately 0.38 in the early 1990s to approximately 0.26 by 2019 — reflecting Colombia's trade diversification toward China, the European Union, and other partners. Voting similarity, by contrast, shows no corresponding trend. If trade were driving alignment, some co-movement between the two series would be expected; its absence is consistent with the null correlation.

Several limitations of this analysis should be noted. The sample consists of a single dyad observed over 29 years. This design is illustrative rather than representative: findings

cannot be generalized to other country pairs or asymmetric relationships. The small sample size also limits the statistical power of the correlation coefficient. Additionally, the analysis is purely descriptive and makes no claim about causation. Omitted variables — including US aid flows, regional bloc membership, ideology of the Colombian government and the US administration, and bilateral security agreements — may confound the relationship in ways that cannot be addressed within a bivariate framework.

Figure 1 presents the full historical trend of Colombia–US voting similarity from 1946 to 2019, extending well beyond the study period. This longer series is included to provide context: it shows that the low level of alignment observed after 1991 is not an anomaly but part of a long structural decline that began in the 1960s and stabilized in the mid-1980s. The study period starts in 1991 for a purely practical reason — bilateral trade data from the World Bank WITS portal is not available for this dyad before that year. It is likely that trade salience was higher in earlier decades, as Colombia's trade was less diversified before the rise of China and other partners as major trade destinations; had earlier data been available, the observed downward trend in trade salience would probably appear longer and more gradual.

5 Conclusion

This paper examined whether the salience of the Colombia–United States trade relationship is associated with Colombia–US voting similarity in the United Nations General Assembly over the period 1991–2019. Trade salience was operationalized as the share of Colombia's total trade conducted with the United States; voting similarity was measured as the annual share of UNGA roll-call votes on which both countries cast identical positions. The bivariate analysis, based on 29 dyad–year observations, yields a Pearson correlation of $r = -0.042$, providing no support for the hypothesis that higher trade dependence is associated with greater political alignment.

The null finding is substantively interpretable. It suggests that, at least in this dyad and period, trade salience does not function as a lever that translates commercial dependence into political conformity at the multilateral level. Political, security, and ideological factors — which varied considerably across Colombian governments during the study period — appear to be more plausible drivers of voting behavior than trade structure.

Future research could address the limitations of this analysis in several ways. Extending the study to a comparative sample of asymmetric dyads would allow for assessment of whether the null result generalizes or is specific to the Colombia–US case. Disaggregating UNGA votes by issue area, and incorporating additional covariates such as US aid, bilateral security agreements, and government ideology, would provide a more complete test of the trade–alignment hypothesis within a multivariate framework.

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Eigenständigkeitserklärung

Ich versichere, dass ich die vorgelegte Seminararbeit eigenständig verfasst, keine anderen als die angegebenen Quellen und Hilfsmittel verwendet und die ihnen entnommenen Passagen als solche kenntlich gemacht habe. Alle verwendeten Hilfsmittel sind in der untenstehenden Tabelle aufgelistet. Bei der Verwendung von Large Language Models (LLM, z.B. Chat-GPT) habe ich den gesamten LLM-Interaktionsverlauf mit Bezug zur vorgelegten Arbeit archiviert und kann ihn auf Verlangen vorzeigen. Diese Seminararbeit ist in dieser oder einer ähnlichen Form in keinem anderen Kurs vorgelegt worden.

Name des Hilfsmittels	**Verwendungszweck**
Claude (Anthropic)	Fehlersuche und Formatierung des Codes gemäß dem Beispielmaterial sowie abschließende Überprüfung der Einhaltung der Kursanforderungen. Der Interaktionsverlauf ist über das Claude-Projekt archiviert.[1]
ChatGPT (OpenAI)	Erstellung und Fehlerbehebung des anfänglichen LaTeX-Codes in Overleaf sowie Fehlerbehebung des anfänglichen R-Codes. Der Interaktionsverlauf kann nicht vorgelegt werden, da das Konto während des Projektzeitraums gekündigt wurde und die Gespräche nicht zuvor archiviert wurden.

Archivierte Interaktionsverläufe

Der Interaktionsverlauf mit Claude (Anthropic) ist über das folgende Claude-Projekt archiviert: `https://claude.ai/project/019cb3e3-677f-7505-a16a-3c1e6add66dc`

Der zugehörige R-Code und alle Projektdateien sind im folgenden GitHub-Repository verfügbar: `https://github.com/santiago2108/aqr-ir-project`

Der Interaktionsverlauf mit ChatGPT (OpenAI) kann nicht vorgelegt werden, da das Konto während des Projektzeitraums gekündigt wurde und die Gespräche nicht zuvor archiviert wurden.

Ort, Datum

Unterschrift